

Introduction to programming

II Mid Question Bank

1. a) Write a program for multiplication of two Matrices
b) Demonstrate the procedure for using the two-dimensional arrays.
2. a) write a program to display whether a given matrix is symmetric or not
b) what is a string? Explain the declaration, initialization, reading and displaying the string
3. a) write a program for copying one string to another without using string handling functions
b) write a program to reverse a string without using string handling functions
4. a) Explain any five string handling functions
b) Explain array of strings? Write a program for sorting array of strings.
5. A) Explain the elements of user defined function.
b) What is a recursion? Write a recursive programs for GCD and Fibonacci series
6. a) Explain the categories of functions based on input parameters and return type
b) Explain the lifetime, scope and longevity of variables (or) storage classes
7. a) What is a pointer? Explain the advantages of the pointers
b) Show the strategy of call by value and call by reference.
8. A) Explain how pointers are used to represent arrays? Write a program for sorting of arrays using pointers
b) Explain dynamic memory allocation in C
9. a) What is a structure? Explain the declaration, initialization and accessing of structure members
b) Explain about nested structures and their usage with an example.
10. a) Explain the differences between structures and unions
b) Explain the ways in which structures can be passed to the functions
11. a) What is file? Explain the operations used on files
b) write a program to copy the contents of one file to another
12. a) Write a C program to read two file names using command line arguments and merge the content of second file to first file.
b) List and explain different functions used for random operations on files.

Short Answer questions

1. what is a function ? What are the advantages of functions

A function is a self contained block or set of instructions written to perform a well defined task

Advantages of functions

- a. code reusability
 - b. modularity
 - c. simplicity and readability
 - d. Ease of maintenance and debugging
 - e. Avoids redundancy
2. Differentiate library functions and user defined functions

Aspect	In-built Library Function	User-Defined Function
Definition	Provided by language/library	Created by the programmer
Usage	Ready to use	Need to be defined before use
Purpose	General-purpose tasks	Specific tasks for the program
Examples	sqrt(), printf()	calculateSum(), displayData()

3. What is a recursion

Recursion is a process of calling a function by itself.

Every recursive function should have a base case and recursive case

4. Write the syntax for declaring a function.

functiontype functionname(list of parameters);

5. Differentiate static and dynamic memory allocations.

Static Memory Allocation	Dynamic Memory Allocation
In the static memory allocation, <u>variables</u> get allocated permanently, till the program executes or function call finishes.	In the Dynamic memory allocation, the memory is controlled by the programmer. It gets allocated whenever a malloc() is executed gets deallocated wherever the <u>free()</u> is executed.

Static Memory Allocation	Dynamic Memory Allocation
Static Memory Allocation is done before program execution.	Dynamic Memory Allocation is done during program execution.
It uses <u>stack</u> for managing the static allocation of memory	It uses heap (not heap data structure) of memory for managing the dynamic allocation of memory
It is less efficient	It is more efficient

6. What is the difference between strcmp() and strncmp()?

The fundamental difference lies in the number of characters compared. strcmp compares the entire strings until a null terminator, while strncmp allows you to specify a maximum number of characters (n) to compare, providing a bounded comparison. This makes strncmp particularly useful for scenarios requiring partial string comparisons

7. What is a pointer? What are the advantages of pointers

Pointer is a variable which points to the address of another variable

Advantages of pointers

- It efficiently handles arrays and strings
- Pointers are used to return multiple values from a function
- Pointers increase the speed of the program
- Pointers decrease the length and complexity of the program
- Pointers are used for dynamic memory allocation
- Pointers are used to create data structures like linked list, stacks, queues etc.,

8. What is \0 in string?

\0 represents the null character or null terminator. Its primary use is to mark the end of a string stored in a character array.

9. Explain pointer arithmetic

Pointer Arithmetic is the set of valid arithmetic operations that can be performed on pointers. there are only a few operations that are allowed to perform on Pointers in C language. These operations are:

- a. Increment/Decrement of a Pointer
- b. Addition of integer to a pointer
- c. Subtraction of integer to a pointer
- d. Subtracting two pointers of the same type
- e. Comparison of pointers

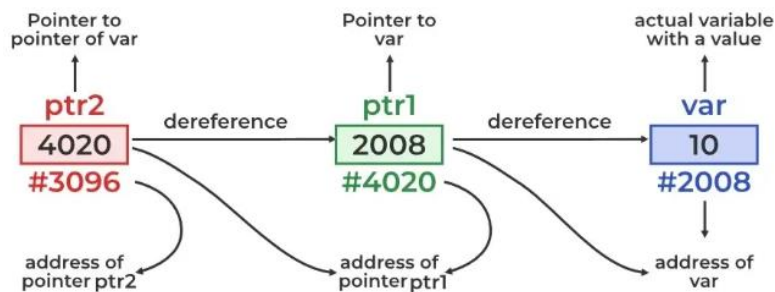
10. What is a dereferencing operator?

the dereference operator, also known as the indirection operator, is represented by the asterisk symbol *. Its primary purpose is to access the value stored at the memory address pointed to by a pointer variable.

11. What is a pointer to the pointer

double pointers are those pointers which stores the address of another pointer. The first pointer is used to store the address of the variable, and the second pointer is used to store the address of the first pointer. That is why they are also known as a pointer to pointer.

```
int var = 10;
int *ptr1 = &var;
// Pointer to pointer (double pointer)
int **ptr2 = &ptr1;
```



12. What are the differences between structure and array

Array	Structure
Stores elements of the same data type (homogeneous collection)	Groups variables of different data types (heterogeneous collection)
Elements are allocated memory in contiguous blocks.	Members are allocated memory separately based on their type
Elements are accessed using numeric indices	Members are accessed using their names with the dot operator
Best for managing a collection of similar data items	Ideal for representing a real-world entity with multiple, potentially varied, attributes

13. Write an example for array of structures

```
struct Student {  
    int rollNumber;  
    char name[50];  
    float marks;  
}s[10];
```

14. What is a self referential structure

self-referential structure is a data structure that contains a member pointing to another instance of the same structure type. This allows for creating dynamic data structures like linked lists, trees, and graphs, where each element, or "node," can be connected to others of its kind.

```
struct node  
{  
    int data;  
    struct node * next;  
};
```

15. How to initialize a structure variable

Initialization using Assignment Operator

You can initialize structure members individually using the assignment operator (=) after declaring the structure variable. Here's an example:

```
struct Point {
```

```

    int x;
    int y;
};

int main() {
    struct Point p1; // Declare a structure variable
    p1.x = 5;
    p1.y = 10;
}

```

Initialization using Initializer List

You can also initialize structure members using an initializer list when declaring the structure variable. Here's an example:

```

struct Point {
    int x;
    int y;
}p1 = {5, 10};

```

16. What is a file? What are the types of the files

File is a collection of data stored permanently in the secondary storage device.

There are two types of files in C

1. Text file
2. Binary File

17. What are the different file modes and which one will be used to create a new file?

File modes specify how a file will be accessed when opened using the `fopen()` function.

These modes determine whether you can read, write, or append to a file, and also whether it's a text or binary file.

Common File Modes in C:

r- read mode

w-write mode

a-append mode

r+ - read and write mode

w+ - read and write mode

a+ - read and append mode

rb,wb,ab,r+b,w+b,a+b are used for binary file

w mode is used to create a new file

18. What is the difference between scanf() and fscanf()

scanf: Reads formatted input from the standard input stream (stdin), which is typically the keyboard.

```
int scanf(const char *format, ...);
```

fscanf: Reads formatted input from a specified file input stream, which is represented by a FILE * pointer.

```
int fscanf(FILE *stream, const char *format, ...);
```

19. What is the purpose of fseek()

fseek function sets the file position of the stream to the given offset.

Syntax

```
int fseek(FILE *stream, long int offset, int pos);
```

FILE *stream: A pointer to a FILE object that identifies the stream.

long int offset: The number of bytes to move the file pointer.

pos: The position from which the offset is added. It can have one of the following values:

SEEK_SET: Beginning of the file.

SEEK_CUR: Current position of the file pointer.

SEEK_END: End of the file.

20. What are command line arguments

Command-line arguments are the values given after the name of the program in the command-line shell of Operating Systems. Command-line arguments are handled by the main() function of a C program.

To pass command-line arguments, we typically define `main()` with two arguments: the first argument is the number of command-line arguments and the second is a list of command-line arguments.

Syntax

```
int main(int argc, char *argv[])
```